

Proposed presentation map — review-weighted talk (for review before slide edits)

Status note. A full 18-slide Beamer deck already exists in `main.tex` (with compiled `main.pdf` / `main-notes.pdf`), written before this map was approved. That deck is a **two-project status talk**: 2 background slides, 6 copper slides, 7 RFX slides. This map re-plans it as a **review talk** that develops background and past results, centers on the copper (cond26) result, and treats RFX as a hint at the future. **No edits to `main.tex` should be made until this map is approved.**

Data-gathering status (2026-07-20) — complete

- **Figures:** all missing cond26 figures copied into `figures/` (see the figure section below). Slides 8 and 11 no longer need placeholders.
- **Sources:** 11 backing PDFs collected into `sources/` — the three Engelberg MDDF papers (PRL 2018, PRAB 2019, PRAB 2020), Jacewicz JAP 2025, the cond26 manuscript PDF, the MeVArc 2025 deck, Wuensch RMP 2026, Nordlund-Djurabekova 2012, Korsback 2020, and the group’s MeVArc 2024 + 2013 decks. Full mapping in `knowledge/source_inventory.md` (“Collected source files” section). **Not found:** Degiovanni PRAB 2016, Mughrabi 2009, Stanzl-Tschegg 2007 (cited from `../cond26/refs.bib` only).
- **Status refresh:** Q14–16 answered in `knowledge/status_refresh_2026-07-20.md` — no new copper data; RFX #2-25/ #3-25 unconfirmed after 13 Jul; cond26 submitted to PRAB 2026-06-23.
- **Not done on purpose:** no edits to `main.tex` and no compile — awaiting the user’s answers in `QUESTIONS.md`.

Format decisions (confirmed 2026-07-19)

- **Length:** 30 minutes **excluding** questions → room for ~16–18 core slides.
- **Balance:** review-weighted — about **30% background / 45% copper / 15% limits+future**, versus the current deck’s copper/RFX co-headline.
- **RFX on the main path: two slides** — one “future directions” slide and one stainless-steel preview. All RFX electrode/SEM/EBSD/artifact detail moves to backup.
- **Closing claim strength: candidate structural basis for E_S** — the article’s careful wording, not “we measured E_S.”
- **E_S is delayed, not a background pillar.** The narrative spine is the group’s own program — the MDDF dislocation model, the local Uppsala TEM microscopy, then the large-area cond26 test. E_S appears only once, late, in the interpretation, as a secondary phenomenological correlate.
- **Two prior-work links are strengthened:** (1) the MDDF dislocation-dynamics model (Engelberg & Ashkenazy, PRL 2018 / PRAB 2019) whose central prediction cond26 tests; (2) the prior microscopy (Jacewicz et al., Uppsala/FREIA, JAP 2025) whose local denuded-zone observation cond26 extends to large area.
- **This talk is the explicit sequel to MeVArc 2025.** The 2025 HUJI talk (“Observing plastic evolution related to high-field conditioning”) presented the MDDF model and the local STEM/FIB microscopy, and closed by asking “*much more comparative forensics — other materials? more samples?*” cond26 is the large-area answer; RFX answers “other materials?”. Use a “last year / this year” callback as the Part I → Part II bridge and as the closing bookend.

Continuity with MeVArc 2025 (source: `mevarc25_huji_v2.pdf`)

The audience largely overlaps with last year’s, so build on the 2025 narrative rather than re-deriving it:

- **2025 spine:** MDDF critical-transition model → model-vs-observation (rate vs field at $T = 100/400$ K; dark-current hypoexponential spikes) → post-BD SEM/plasma deposition → sub-surface predictions → STEM/FIB microscopy → denuded zones in the top ~100 nm of conditioned Cu (300 K and 30

K) → field-exposed vs reference comparisons → “method and its limits established; what next? other materials? more samples?”

- **2025 assets worth reusing** (embedded in .pptx, extract if approved): the MDDF governing-equation/critical-transition schematic; the rate-vs-field fit curves; the dark-current spike distribution; representative cross-sectional STEM denuded-zone panels; the FIB “red herring” artifact examples.
- **2025 lesson to carry into RFX:** FIB sample production generates dislocation artifacts (“red herrings”) that masquerade as structure; large-scale features survive but fine structure does not. This is first-hand justification for the RFX depth-artifact controls.
- **Number to reconcile:** the 2025 talk says the modification is limited to the **top ~100 nm**, while cond26/main.tex and Jacewicz JAP 2025 quote **~200 nm** (and Groma $\rho_D \sim 100$ nm). Pick one consistent figure for 2026.

Proposed communication job

By the end, a MeV Arc audience should understand (1) that the group’s MDDF model predicts conditioning is collective mobile-dislocation evolution driven by the Maxwell stress, (2) that prior microscopy (the Uppsala TEM study) saw the predicted structural change but only locally, (3) that cond26 provides the first large-area structural confirmation — a field-correlated subsurface misorientation signature in copper — and (4) how a depth-resolved and stainless-steel program tests whether the effect is universal.

Central message

Model (prior): The mobile-dislocation-density fluctuation (MDDF) model of Engelberg & Ashkenazy treats breakdown initiation and conditioning as collective dislocation dynamics driven by the tensile Maxwell stress, and reproduces the E^{30} breakdown-rate scaling and dark-current behaviour. Its central prediction is that conditioning evolves the near-surface dislocation structure.

Microscopy (prior, local): Cross-sectional TEM of conditioned copper (the Uppsala/FREIA study, Jacewicz et al., with Popov and Ashkenazy) found a local ~200-nm dislocation-denuded zone — direct but single-site structural evidence.

New result (center): Large-area EBSD shows high-field copper regions with substantially higher intragrain misorientation than low-field and unexposed controls, in a three-tier spatial hierarchy with a broadened distribution — the missing large-area structural test.

Interpretation (careful, E_S delayed): The result is the first large-area confirmation of the MDDF prediction and is consistent with accumulated, field-driven dislocation activity; the spatial ordering also tracks the phenomenological conditioning variable E_S , making the evolving subsurface dislocation population a **candidate structural basis** for E_S — not a proven identity.

Boundary: One conditioned cathode; causality, depth profile, and universality are not yet established.

Audience takeaway

The dislocation picture of conditioning now has direct large-area structural support, not just a model and a single TEM cross-section. The next decisive experiments connect lateral exposure to depth and test a material/loading contrast.

Recommended balance and arc

- Core talk: about 16–18 slides, ~28 minutes; questions after the slot.
- Arc: operational problem → MDDF model and its prediction → prior local microscopy and its limit → why a sub-yield Maxwell stress can matter → controlled copper experiment → visual observation → quantitative hierarchy → distribution stretch → confirmation of the model (+ delayed E_S note) → limits → depth and cross-material future.
- Reuse the existing deck’s copper slides and visual conventions; the main work is **re-spinning Part I around MDDF + microscopy and shrinking RFX to two slides**.

Extended introduction for the expected audience

Assume familiarity with vacuum breakdown and conditioning; do not assume EBSD or dislocation-screening background. This is a genuine review of the group's program, so the introduction builds the model-then-microscopy case before the new data. E_S is deliberately withheld until the interpretation.

1. Operational fact: conditioning sets the usable field; breakdown rate scales roughly as E^{30} ; conditioning tracks high-field **pulse count** more than breakdown count, implying the field itself drives conditioning, not arc damage.
2. The model: FCC/BCC/HCP field-holding ordering $\sim$ barriers to dislocation motion; the MDDF model treats breakdown/conditioning as collective mobile-dislocation dynamics under the Maxwell stress and reproduces the E^{30} and temperature trends and dark-current behaviour. **Its central prediction — conditioning evolves the near-surface dislocation structure — is the hypothesis this talk tests.**
3. The prior microscopy: the Uppsala TEM study found a local ~ 200 -nm denuded zone in conditioned copper, plus the hard-vs-soft conditioning contrast — direct structural evidence, but local and hard to extrapolate.
4. Why a tiny Maxwell stress can matter: $\sigma_M = E^2/2 = 0.028$ MPa, $\sim 10^3$ below yield, but VHCF shows sub-yield cycling over $\sim 10^7$ pulses accumulates irreversible rearrangement, and the MDDF description is threshold-free.
5. Name the precise missing experiment: a **large-area** structural comparison across controlled exposure on one electrode — the gap the model and the local microscopy both leave open.
6. Teach only the EBSD concept needed: low-angle intragrain orientation gradients proxy GND/plastic activity; grain boundaries are excluded.

Suggested spoken transition into results (the 2025 $\rightarrow$ 2026 bridge):

Last year I showed this model and our STEM cross-sections, and ended by asking for more comparative forensics — other materials, more samples. This year we answer part of that: a full-electrode map that should know where the field was high, even far from breakdown craters.

Slide-by-slide narrative

Part I — A model, local microscopy, and the missing large-area test (~ 4 slides)

1. Title: “Does conditioning leave a structural memory?”

- Job: pose the scientific problem; signal a review that lands on a new result.
- Visual: restrained title; existing title frame is reusable.

2. “Conditioning works — but the mechanism is unknown”

- Highlight: conditioning raises voltage holding; steep E^{30} breakdown-rate scaling; conditioning tracks high-field **pulse count**, not breakdown count $\rightarrow$ the field, not arc damage, drives conditioning.
- Evidence: Introduction lines 52–53, 66; Wuensch et al. RMP 2026; Degiovanni et al. PRAB 2016.
- **Do not introduce E_S here** (delayed to Slide 12).
- Reuse: current Slide 2 left column and the exposure/field-holding sketch; drop its E_S box.

3. “Our model: conditioning as collective dislocation dynamics (MDDF)”

- **Strengthened prior-work link (model).** Field-holding order FCC $\rightarrow$ BCC $\rightarrow$ HCP tracks barriers to dislocation motion; the thermodynamic defect model (Nordlund–Djurabekova) couples field stress to near-surface loop formation.
- The **MDDF model** (Engelberg, Ashkenazy et al.): breakdown initiation as a critical transition in collective mobile-dislocation motion driven by the tensile Maxwell stress; reproduces the E^{30} scaling and temperature dependence; pre-breakdown dark-current spikes are consistent with it.
- **Central prediction to foreground:** conditioning evolves the near-surface dislocation structure — the hypothesis cond26 tests.

- Reusable 2025 figures (already copied to `figures/`, see `knowledge/mevarc25_assets.md`): `mddf_critical_transition.png` (the dn/dt vs n runaway schematic), `mddf_rate_vs_field.png` (the steep $\sim E^{30}$ scaling), and `mddf_nucleation_rate.png` (rate vs $E - E_{th}$).
- Evidence: Introduction lines 56–61; engelberg_stochastic_2018 (PRL 120, 124801, 2018); engelberg_theory_2019 (PRAB 22, 083501, 2019); engelberg_dark_2020 (PRAB 23, 123501, 2020); nordlund_defect_2012; serafim_effects_2025.
- Replaces the current deck’s E_S slide.

4. “Prior microscopy saw the structure — but only locally (Uppsala TEM)”

- **Strengthened prior-work link (microscopy) — this is our own 2025 STEM work.** Cross-sectional STEM/TEM of conditioned copper (Jacewicz & Profatilova, Uppsala/FREIA; Popov, HUJI; with Ashkenazy) revealed a dislocation-denuded zone in the top ~ 100 – 200 nm of field-exposed regions (both 300 K and 30 K) — the first *direct* structural evidence of field-driven near-surface plasticity. Reusable figures already in `figures/`: `stem_denuded_zone.png` (denuded band, 100/200 nm scale bars) and the `stem_fe_300k.png` / `stem_ref_300k.png` field-exposed-vs-reference pair.
- Support: soft (heat-treated) copper conditions more slowly than hard (as-machined) copper, consistent with different initial dislocation populations.
- **Carry the 2025 “red herring” caution:** FIB preparation itself generates dislocation artifacts; only large-scale structure survives. This both scopes the microscopy claim and pre-motivates the RFX artifact controls.
- **Limit that motivates cond26:** STEM/TEM is highly local and hard to extrapolate to the full electrode $\rightarrow$ the large-area structural comparison was still missing; the 2025 talk closed on exactly this (“other materials? more samples?”).
- Fold the Maxwell-stress/VHCF point ($\sigma_M = 0.028$ MPa, ~ 10 sub-yield cycles, threshold-free MDDF) into this slide or a compact half-slide, so Part I stays at ~ 4 slides.
- Evidence: Introduction lines 63–71, 113; jacewicz_surface_2025 (JAP 137, 193302, 2025); korsback_vacuum_2020; mughrabi_cyclic_2009; stanzl-tscheegg_vhcf_2007; MeVArc 2025 deck.
- **New/strengthened slide** relative to the current deck.

Part II — The copper result (center, ~ 7 slides)

5. “One cathode turns field exposure into a controlled coordinate”

- Figure: Fig. 1 geometry (`electrode_geometry.png`); optionally Fig. 3 field profile (`efield_vs_radius.eps`, needs copying).
- Highlight: 80 MV/m center; ~ 69 – 77 MV/m edge; low-field periphery; one cathode plus a separate unexposed reference.
- Reuse: current Slide 4.

6. “EBSD maps plastic activity far from breakdown craters”

- Figure: Fig. 2a (`sem_lowmag.png`).
- Highlight: ROIs hundreds of microns from craters; identical cleaning; LAM primary, KAM/LOS as cross-checks; misorientation as GND proxy; EBSD chosen precisely because TEM is only local.
- Evidence: Methods lines 113, 177–189.
- Reuse: current Slide 5.

7. “How to read a LAM map: wide-angle vs low-angle”

- Figure: Fig. 5 pair — `lam_widescale.png` (0 – 50°) and `lam_lowscale.png` (0 – 5° , needs copying).
- Job: teach the metric before the comparison.
- **Relocated slide.** In the current deck `lam_widescale.png` is used on the distribution slide, which is a mismatch to fix.

8. “High-field copper contains more intragrain curvature”

- Figure: `lam_fe_center.png` vs `lam_ref_edge.png`, same $0\text{--}5^\circ$ scale.
- Highlight: qualitative spatial observation before statistics.
- Reuse: current Slide 6.

9. “The signal separates into three exposure tiers”

- Figure: Fig. 9 (`mean_vs_radius.pdf`).
- Numbers: $\sim 1.2^\circ$ high field; $\sim 0.79^\circ$ periphery; $\sim 0.68^\circ$ external REF; $\sim 75\%$ difference; grain size does not follow the hierarchy.
- Primary result slide. Reuse: current Slide 7.

10. “Field exposure populates a high-misorientation tail”

- Figure: Fig. 8 (`low_angle_overlay.pdf`, needs copying) — **replace** the current wide-scale LAM image here.
- Numbers: $P(\text{LAM} > 2^\circ) \sim 0.14$ vs ~ 0.016 ($\sim 8\times$); common gamma $k = 2.7$, scale roughly doubles; $D_{\text{KS}} > 0.25$ against all unexposed distributions.
- Caveat: emphasize effect size and grain-corrected errors, not tiny correlated-pixel p-values. Reuse: current Slide 8 (swap figure).

11. “The large-area test confirms the MDDF prediction”

- **Payoff slide — closes the loop to Part I.** Lead: `cond26` is the first large-area confirmation that conditioning evolves the near-surface dislocation structure, the central MDDF prediction. Chain: field $\rightarrow$ Maxwell stress (~ 10 cycles) $\rightarrow$ dislocation rearrangement $\rightarrow$ measured LAM hierarchy.
- Connect to microscopy: EBSD **extends the local Uppsala TEM observation** to the full conditioned region; the surface denuded zone and the deeper accumulation are reconciled at different depths (Groma screening length $\lambda_D \sim 100$ nm matches the ~ 200 -nm zone).
- **Delayed E_S note (secondary):** the three-tier spatial ordering also follows the phenomenological $E_S(\mathbf{r})$ profile from the conditioning simulations, so the dislocation population is a **candidate structural basis** for E_S . Keep this to one line; do not say “we measured E_S .”
- Evidence: Introduction lines 61, 80; Discussion lines 369–385, 411–427.
- Reuse: current Slide 9, re-led on MDDF rather than on E_S .

(Optional) “The annealed grain skeleton stays intact”

- Figure: Fig. 10 (`full_range_overlay.pdf`, needs copying) or Fig. 4 IPF pair.
- Content: $\Sigma 3$ fraction unchanged; change is intragrain/subgrain. Say the mid-angle excess is **consistent with** continued subgrain rotation (per the `TODO.md` wording concern).
- **Removable** if the talk runs long (first to cut).

Part III — Limits and hints at the future (~ 4 slides, two RFX)

12. “Copper establishes the effect, not universality”

- Split: established (large-area difference; three metrics; internal + external controls; confirms the MDDF prediction; ordering consistent with E_S) vs open (depth profile; grain-orientation dependence; more cathodes; cross-material generality).
- Transition slide. Reuse: current Slide 10.

13. “Two next coordinates: depth and a change of material”

- **RFX slide 1 of 2 (future directions).** Merge the depth program and the cross-material motivation.

- Concrete depth plan: matched high-field/reference pilot FIB lamellae; identical artifact controls (informed by the 2025 “red herring” experience); low-energy final polish; correlated TEM/STEM + EBSD through the top $\sim 1\ \mu\text{m}$ — does the $\sim 100\text{--}200\text{-nm}$ surface depletion join the deeper accumulation into one redistribution profile? (Directly continues the 2025 Uppsala STEM line.)
- Cross-material motive: stainless steel changes stacking-fault energy, dislocation mobility, and loading mode.
- Condenses current Slides 11 and 16; artifact detail goes to backup.

14. “Stainless steel: a deliberately difficult, falsifiable test”

- **RFX slide 2 of 2 (preview).** AISI 304, lower SFE, metastable austenite, continuous DC, two exposed + two matched references, apex-to-periphery gradient.
- Falsifiable outcomes: field-correlated SS response supports generality; a different response identifies material-specific carriers; no resolved response bounds depth/loading/sensitivity.
- Keep preliminary SEM/EBSD observations **in backup**, labeled and only if approved. Condenses current Slides 12–15, 17.

15. “Conditioning appears to leave a subsurface memory”

- Recap along the program spine: MDDF predicted structural evolution; the 2025 Uppsala STEM saw it locally; cond26 confirms it large-area ($\sim 75\%$); depth and stainless steel test whether the memory survives a change of length scale, alloy, and loading. One-line **E_S** candidate close.
- **Bookend the 2025 talk:** last year closed on “other materials? more samples?”; this year delivers the large-area copper answer and hands off to the stainless-steel/depth program — the next round of comparative forensics.
- Reuse: current Slide 18; `infographic.pdf` optional.

Figures to copy from `cond26/figures/` into `mevarc26/figures/`

DONE (2026-07-20). All copied into `figures/`:

- `low_angle_overlay.pdf` (Fig. 8) — real distribution/tail plot for Slide 10.
- `lam_lowscale.png` (Fig. 5b) — pair with `lam_widescale.png` for Slide 7.
- `efield_vs_radius-eps-converted-to.pdf` (Fig. 3) — optional field profile for Slide 5 (already PDF; no eps conversion needed).
- `full_range_overlay.pdf` (Fig. 10) and IPF maps (`ipf_fe_center.png`, `ipf_ref_center_keyed.png`, `ipf_key.png`) — available if the optional grain-skeleton slide is kept.

Results deserving the most emphasis

1. Confirmation of the MDDF prediction by a large-area structural measurement.
2. Fig. 9 three-tier hierarchy and $\sim 75\%$ effect.
3. Fig. 8 distribution stretch and $\sim 8\times$ tail.
4. Agreement of LAM, KAM, and LOS.
5. Internal-control logic: same cathode, low-field periphery, ROIs away from craters; continuity with the Uppsala TEM cross-sections.

Limitations and alternative interpretations to keep visible

- One conditioned cathode and one external reference.
- Field level and breakdown density are not independently varied.
- Pixel-pair p-values overstate independent sample count; report effect sizes and grain-corrected errors.
- LAM is a relative GND proxy at fixed step size, not an absolute density.
- FIB/indexing artifacts controlled by identical protocol, not exhaustively eliminated.
- Depth reconciliation and **E_S** identity are hypotheses.
- Soften the mid-angle/subgrain statement pending confirmation.

Concrete versus speculative future directions

Concrete

- matched high-field/reference pilot lamellae; artifact floor and low-energy final polish; depth-resolved TEM/STEM through $\sim 1\ \mu\text{m}$ (continuing the Uppsala line); finer radial EBSD for quantitative $E_S(\mathbf{r})$; additional cathode replication; RFX field-gradient EBSD with phase mapping and matched references.

Speculative (one sentence, close or backup)

- pulse-spectrum coupling to dislocation resonances; a cross-material/loading “phase diagram” of conditioning response; extension to RF, cryogenic, BCC, and HCP systems.

Optional/removable slides if the talk runs long

1. Optional grain-skeleton / $\Sigma 3$ slide (first to cut).
2. Merge Slides 6–7 (EBSD method + how-to-read-LAM) into one.
3. Fold Slide 12 caveats into Slides 9, 10, and 11.
4. Backup only: RFX status table, SEM survey, early EBSD pattern quality, depth-artifact controls, exact Table I values, pairwise-test matrix.

Open scientific questions for review

1. Is the MDDF “central prediction” framing (Slide 3 $\rightarrow$ Slide 11) the spine you want, with E_S only as a one-line correlate at Slide 11/15?
2. **Denuded-zone depth:** resolved — `stem_denuded_zone.png` carries both 100 nm and 200 nm scale bars (different magnifications of the same layer), so 2026 can say “top $\sim 100\text{--}200\ \text{nm}$.” Confirm you are happy with that phrasing.
3. How explicit should the 2025 callback be — a light spoken “last year/this year” bridge, or a dedicated recap slide at the Part I $\rightarrow$ Part II boundary?
4. Figure reuse: six 2025 figures are already copied into `figures/` (see `knowledge/mevarc25_assets.md`). Confirm they may be used; say if you want the FIB “red herring” or post-BD SEM panels pulled in too.
5. Do you endorse the hard/soft copper depth-reconciliation model (Groma `_D`) for a main-path sentence on Slide 11, or keep it in backup?
6. Should the optional grain-skeleton slide stay in the 30-minute cut?
7. Has newer data changed the one-cathode limitation or RFX status? **Answered 2026-07-20** (`knowledge/status_refresh_2026-07-20.md`): no new copper data (one-cathode caveat stands); RFX last record 13 Jul — #1-25 EBSD good, #4-25 poor, #2-25/#3-25 scheduled 14 Jul with no record since; cond26 submitted to PRAB 2026-06-23, no reviewer changes.
8. Is there a quantitative EBSD information-depth estimate to replace the qualitative “deeper than the TEM surface zone” language?

Open presentation-design questions for review

1. Audience mix: breakdown specialists, materials scientists, or both?
2. Confirm E_S stays out of Part I entirely and appears only at Slide 11/15.
3. May approved preliminary RFX SEM/EBSD appear even in backup, or is RFX strictly a clean future-direction preview?
4. Confirm the author/affiliation block. From the 2025 title slide: Ashkenazy (HUJI/Racah); Profatilova, Jacewicz (Uppsala); Popov (HUJI); Bjelland, Wuensch, Millar, Calatroni (CERN). Add RFX collaborators for the SS section.
5. Style: keep the current `main.tex` palette/title, or align more closely with the recommendations in `knowledge/style_inventory.md`?

Approval gate

Before editing `main.tex`, please approve or revise:

- the model→microscopy→large-area→future spine, with **E_S** delayed;
- the strengthened MDDF (Slide 3) and Uppsala-TEM (Slide 4) prior-work links;
- the ~16–18-slide Part I / Part II / Part III structure;
- the two-slide RFX treatment (rest to backup);
- whether the optional grain-skeleton slide stays;
- copying the four missing figures listed above.